

Corporate Attribution Ratio of Fossil Fuel CO₂ Emissions Across Countries

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Final Report

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1.0 Abstract

This study examines how production-attributed fossil fuel CO₂ emissions compare with territorial fossil fuel CO₂ emissions across countries from 2000 to 2020. Using data from the Carbon Majors database, the Global Carbon Project, GDP data, and the Regulatory Indicators for Sustainable Energy (RISE), we constructed a Corporate Attribution Ratio defined as production-attributed emissions divided by territorial emissions for each country-year. The ratio was used to evaluate how corporate-linked emissions vary across countries and over time, and whether they are associated with economic and policy characteristics.

Results showed substantial variation in the Corporate Attribution Ratio across countries. Time trends also differed across countries, with some showing significant increases and others significant decreases over the study period. Comparisons of production-based and territorial emissions per capita suggested a statistically significant difference overall, although the magnitude was modest and the distribution was highly skewed. Stronger renewable energy regulatory scores were generally associated with lower Corporate Attribution Ratios and a lower likelihood that production-attributed emissions exceeded territorial emissions. Overall, the findings suggest that renewable energy policy environments may be more informative than economic development alone in understanding cross-country differences in carbon emissions linked to corporations.

2.0 Introduction

2.1 Background

Understanding the sources of global carbon emissions is essential for effective climate policy. Fossil fuel production and consumption account for over 80% of global energy usage (Mohr et al., 2015). While national emissions inventories typically measure territorial emissions, greenhouse gases produced within a country's borders, this analysis focuses on corporate-attributed emissions, which trace responsibility back to the fossil fuel producers extracting and marketing carbon-based fuels.

Research shows that a small number of producers are responsible for a large share of global industrial emissions (Miklin et al., 2024). The Carbon Majors database links production quantities to estimated CO₂

emissions for major fossil fuel and cement producers, enabling an assessment of corporate responsibility regardless of where emissions are physically released. The Global Carbon Project, by contrast, examines emissions on a territorial basis.

To compare these two frameworks, we construct a Corporate Attribution Ratio. Some countries host corporations responsible for large emissions volumes that are released elsewhere, creating potential gaps between production-based and territorial accounting. National factors, including GDP per capita, renewable energy adoption, and regulatory environments, may further shape these differences.

This project integrates multiple global datasets to examine how corporate emissions contributions vary across countries and relate to economic and policy characteristics, exploring how corporate actors influence the trajectory of climate change (Kobie, 2017).

2.2 Report Outline

This report covers data sources, analytical methods, and findings organized around five research questions:

1. How does the Corporate Attribution Ratio vary across countries, and what is its global distribution?
2. How does the Corporate Attribution Ratio evolve within countries?
3. Is there a significant difference between production-attributed fossil CO₂ emissions per capita and territorial fossil CO₂ emissions per capita?
4. Is the Corporate Attribution Ratio associated with renewable energy regulations?
5. What key dimensions drive variation in the Corporate Attribution Ratio across countries?

The report begins with data cleaning and codebook construction, followed by preliminary analysis addressing each question. It then presents results, evaluates how well the analysis meets our research goals, and concludes with key findings, limitations, and their broader implications.

2.3 References

Throughout the course of the report, studies involving global carbon emissions on a country and corporate level were heavily referenced. Heede's foundational work traces CO₂ and methane emissions to fossil fuel and cement producers from 1854–2010, establishing the basis for the Carbon Majors Project and quantifying the extent of corporate responsibility for global emissions (Heede, 2013). At the country level, Andrew's comparison of global CO₂ emission estimates examines coal, oil, gas, and cement metrics within national borders, informing our use of the Global Carbon Project data (Andrew, 2020). Finally,

Freire-Gonzales et al. find a positive but weakening relationship between GDP per capita and CO₂ emissions per capita, categorizing country-level associations into six patterns using logarithmic regression, providing a useful framework for interpreting our own GDP and emissions analysis (Freire-Gonzales et al., 2024).

2.3 Data Sources and Collection

2.3.1 Carbon Majors Data Set

The Carbon Majors database quantifies historical greenhouse gas emissions attributable to the world's largest fossil fuel and cement producers (CarbonMajors, 2024). Developed by Richard Heede in 2013 at the Climate Accountability Institute and now maintained by InfluenceMap, it traces corporate responsibility for industrial emissions dating back to 1854. The 2024 dataset covers 178 major producers, accounting for approximately 70% of global fossil fuel and cement CO₂ emissions since industrialization, and classifies entities as investor-owned, state-owned, or nation-state producers (Heede, 2013; InfluenceMap, 2024).

Emissions are estimated from the production and combustion of marketed fossil fuels and reported in million metric tonnes of CO₂ (MtCO₂). Total emissions combine Scope 1 (operational emissions from powering production facilities) and Scope 3 (upstream and downstream value chain emissions, primarily from the use of sold products). Production data is drawn from company filings and reports, with third-party sources used when unavailable.

A key limitation is that production emissions in the Carbon Majors dataset are aggregated globally and attributed to a corporation's headquarters country, rather than to the country where emissions are physically released. A corporation headquartered in Norway, for example, may extract and sell fossil fuels that are combusted across dozens of countries, yet all associated emissions are credited to Norway. This inflates production figures relative to territorial emissions for major fossil fuel-exporting nations, causing the Corporate Attribution Ratio to exceed 1 in many cases, even when a country's domestic emissions footprint is relatively modest.

This aggregation also limits the granularity of cross-country comparisons. Because Scope 3 emissions, which account for the majority of corporate emissions, reflect the global downstream use of sold products, they capture activity far beyond a headquartered country's borders. As a result, the Corporate Attribution Ratio reflects corporate reach and influence rather than a direct comparison of emissions occurring within the same geographic space. A more granular attribution of production emissions by country of release would allow for a more precise alignment between corporate and territorial accounting,

and would strengthen the interpretability of the ratio as a measure of comparative emissions responsibility.

2.3.2 Global Carbon Project Data

The Global Carbon Project, launched in 2001 under Future Earth, examines how countries contribute to global carbon emissions across biophysical and human dimensions (Global Carbon Project, 2025). The 2025 release, authored by Robbie Andrews and Glen P. Peters at CICERO, extends coverage from 1750 to 2025 with ongoing refinements to improve accuracy (Andrew & Peters, 2025).

The dataset includes country identifiers (name, ISO label, UN code), year, and territorial emissions broken down by source, coal, oil, gas, cement, and flaring, along with total and per capita figures, all measured in MtCO₂. Data is sourced from CDIAC, UN energy statistics, and the Statistical Review of World Energy, building on foundational emissions estimates dating back to Keeling and Rotty (1973) and expanded through Andrew (2020).

2.3.3 Regulatory Indicators for Sustainable Energy

The Regulatory Indicators for Sustainable Energy (RISE) database was used to measure the strength of national sustainable energy policy (RISE, 2025). Developed by the World Bank's Energy Sector Management Assistance Program (ESMAP), RISE evaluates how supportive a country's regulatory framework is for renewable energy, energy efficiency, and energy access.

Countries receive scores from 0 to 100 based on multiple indicators and sub-indicators capturing specific policy features. These are standardized and aggregated into composite scores for each policy area.

This study focuses on the renewable energy score, which reflects policy dimensions such as national targets, pricing incentives, market design, and grid integration. As a composite measure, it provides a broad view of regulatory support rather than any single policy mechanism.

A limitation is that RISE measures policy design rather than implementation, so higher scores do not necessarily correspond to greater renewable energy adoption. Despite this, it remains a useful and standardized measure for cross-country comparison.

Variable	Mean	SD	Min	Max	N
Corporate Attribution Ratio	2.12	2.29	0.03	13.49	991
Production Emissions Per Capita	25	44.86	0.32	292.01	991
Territory Emissions Per Capita	9.58	9.08	0.48	67.73	991
GDP Per Capita (USD)	24,459.55	25,360.04	383.34	12,3514.2	846
Renewable Energy Score	30.4	21.62	0	82	431

Table 1 - Descriptive Statistics on Key Metrics of the Study

3.0 Methods

3.1 Compiled Dataset and Codebook

Data cleaning and merging were conducted in Python and R using four datasets covering 2000–2020.

The Global Carbon Project data was cleaned by removing unnecessary columns, filtering to 2000–2020, standardizing country names, and adding continent labels. A population column was derived by dividing total emissions by per capita emissions (Andrew, 2020). The Carbon Majors data was similarly cleaned and filtered; AI was used to map all 168 corporations to their headquartered country, and production emissions were aggregated by summing corporate emissions within each country per year (InfluenceMap, 2024b). The filtered years represent 43.2% of the full dataset.

$$\text{Corporate Attribution Ratio} = \frac{\text{Production-Based Emissions Year}_i \text{ Country}_j}{\text{Territorial-Based Emissions Year}_i \text{ Country}_j}$$

Production-based emissions include Scope 1 and Scope 3, capturing global emissions resulting from a corporation's activities. Values above 1 indicate that corporations headquartered in a country are responsible for more emissions globally than are physically produced within its borders; values below 1 indicate the opposite.

Additional variables were created for the proportion of corporations headquartered in each country and continent per year, per capita emissions, and the difference between production and territorial per capita emissions (Altan, 2021). A seven-level categorical emissions variable was also constructed. Finally, GDP

per capita was merged from the Global Data on Sustainable Energy dataset (Tanwar, 2023), and regulatory scores were merged from the RISE dataset for 2010–2020 (RISE, 2025).

3.2 Preliminary Analysis

Preliminary analysis explored corporate distribution across countries over time using visualizations and linear regression in R. Given that China, Russia, and the United States are the world's largest carbon emitters, background analysis focused on these countries, with map visualizations illustrating the relationship between production and territorial emissions (Ritchie, 2025).

3.3 Global Distribution of Corporate Attribution Ratio

To examine the global distribution of the Corporate Attribution Ratio, we mapped it using a carefully constructed gradient. Given the distribution's skew, we created a categorical variable with finer granularity at lower values, using the following boundaries:

1. Extremely Low - Corporate Attribution Ratio less than 0.5
2. Very Low - Corporate Attribution Ratio greater than 0.5 and less than 1
3. Low - Corporate Attribution Ratio greater than 1 and less than 2
4. Moderate - Corporate Attribution Ratio greater than 2 and less than 4
5. High - Corporate Attribution Ratio greater than 4 and less than 6
6. Very High - Corporate Attribution Ratio greater than 6 and less than 8
7. Extremely High - Corporate Attribution Ratio greater than 8

For R visualizations, the ratio was split into numerical bounds: 0.2, 0.5, 0.7, 1, 2, 3, 4, and 6. A chi-squared test was used to assess the relationship between continent and categorical ratio, and summary statistics were computed overall and across years.

3.4 Corporate Attribution Ratio with Time

To examine how the Corporate Attribution Ratio changed over time, separate linear regressions were estimated for each country using year (centered at 2000) as the predictor, yielding a slope, p-value, and R^2 per country. Both Pearson and Spearman correlations were computed, the latter to reduce sensitivity to outliers, with country-specific OLS regressions used when the two disagreed. Time-series plots were created for selected countries representing strong positive, strong negative, and non-significant trends.

3.5 Difference in Production and Territory Emissions

To test the difference between production and territorial emissions, we first visualized the distribution, which was right-skewed due to high production outliers. The Shapiro-Wilk test returned a very low p-value, indicating non-normality. We then applied a Wilcoxon signed-rank test, a non-parametric alternative to a paired t-test, which was not statistically significant, as its rank-based approach reduces the influence of outliers. To reconcile this, we applied bootstrapping, which resamples observed differences to build an empirical distribution. This result was statistically significant, though the magnitude of the difference was small.

3.6 Corporate Attribution Ratio and Renewable Energy Regulations (2010 - 2020)

Using country-level observations from 2010 to 2020 (excluding missing values), we examined the relationship between the Corporate Attribution Ratio and renewable energy regulatory strength, measured by the RISE score. Pearson and Spearman correlations assessed linear and monotonic associations, with Spearman used to account for outliers. A binary indicator (1 if ratio > 1, 0 otherwise) was created, and a Welch two-sample t-test compared mean RISE scores between groups. Finally, a logistic regression modeled the probability that corporate emissions exceed territorial emissions as a function of the renewable energy score.

3.7 Country Emissions Profiles via Principal Component Analysis

Principal Component Analysis (PCA) was used on 11 standardized numeric features, including emissions, per-capita emissions, population, renewable energy scores, and more. Standardization was needed before fitting because PCA is sensitive to scale. Without standardization, features with larger magnitudes would dominate the components even if they were unimportant. A full PCA was fit, and the number of components to keep was determined using a scree plot, landing on 3 components. Reducing from 11 features to 3 makes the visualization of the countries much easier to understand.

Because PCA is an unsupervised method, we do not automatically know what each Component is made up of. This can be explained by a heatmap of contributions, looking at which features cluster together and take over each axis.

3.8 Corporate Attribution Linear Regression via Principal Components

To determine if a country's structural emissions profile can predict how its corporate footprint diverges from its territorial one, a log-linear regression model was fit using the 3 Principal Components as

predictors and the log-transformed corporate attribution ratio as the outcome. The corporate attribution ratio, production emissions divided by territorial emissions, was log-transformed before modeling. Diagnostic plots confirmed the raw distribution was heavily right-skewed, but the log-transformed data was much closer to symmetrical. Because Principal Components are orthogonal and uncorrelated, multicollinearity was not a concern. Model fit was assessed using R^2 and adjusted R^2 , as well as using t-tests to check significance. Regression assumptions were checked through residual diagnostics, such as a residuals vs fitted plot, histogram of residuals, Q-Q plot, and a Shapiro-Wilk test on the residuals.

4.0 Results

4.1 Preliminary Research Results

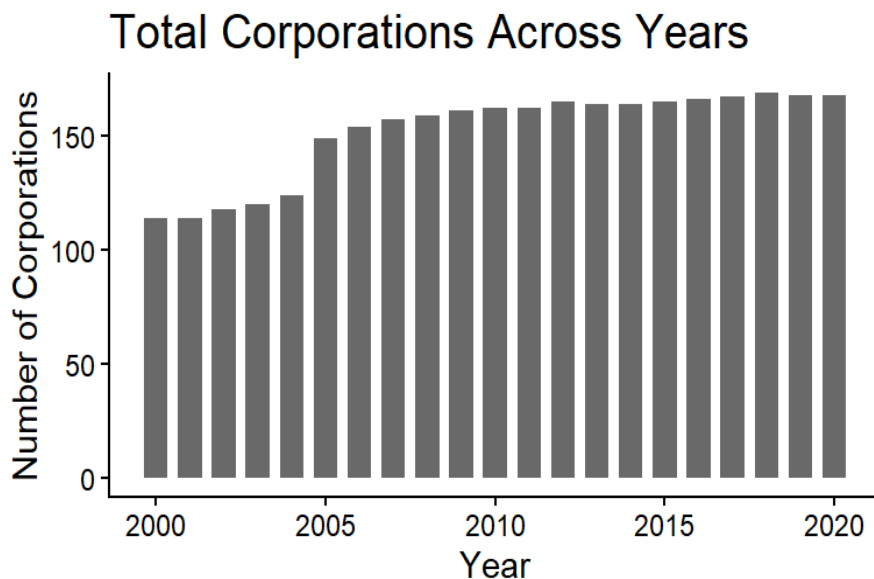


Figure 1 - Number of Corporations from 2000 to 2020

The total number of corporations in the Carbon Majors dataset increased from 114 in 2000 to 168 in 2020. A simple linear regression confirmed a statistically significant positive relationship between year and total corporation count ($p < 0.001$). As shown in Table 2, the

United States, China, and Russia consistently headquartered the largest number of corporations throughout the study period, while all other countries hosted between 1 and 7 corporations in any given year.

Country	Number of Corporations 2000	Number of Corporations 2005	Number of Corporations 2010	Number of Corporations 2015	Number of Corporations 2020
China	5	29	31	31	32

Russia	12	14	17	17	18
USA	24	25	26	28	28

Table 2 - Number of Corporations in China, Russia, and the USA

In 2000, production emissions ranged from 11.73 to 4,650.89 MtCO₂, and territorial emissions ranged from 8.70 to 6,023.16 MtCO₂ globally. By 2020, production emissions ranged from 9.69 to 10,141.14 MtCO₂ and territorial emissions from 8.07 to 10,896.52 MtCO₂. This represents a 43.8% increase in total production emissions and a 40.81% increase in total territorial emissions over the study period (Figures 2.1–2.4).

Figure 2.1

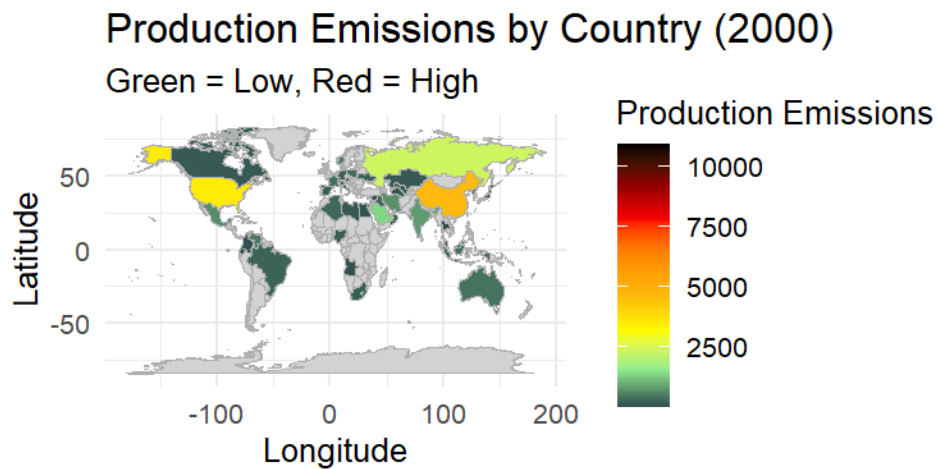


Figure 2.2

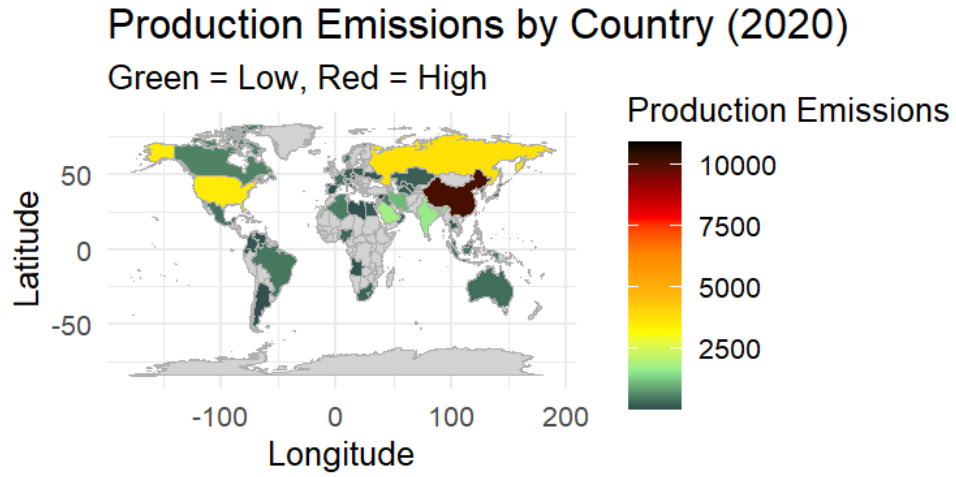


Figure 2.3

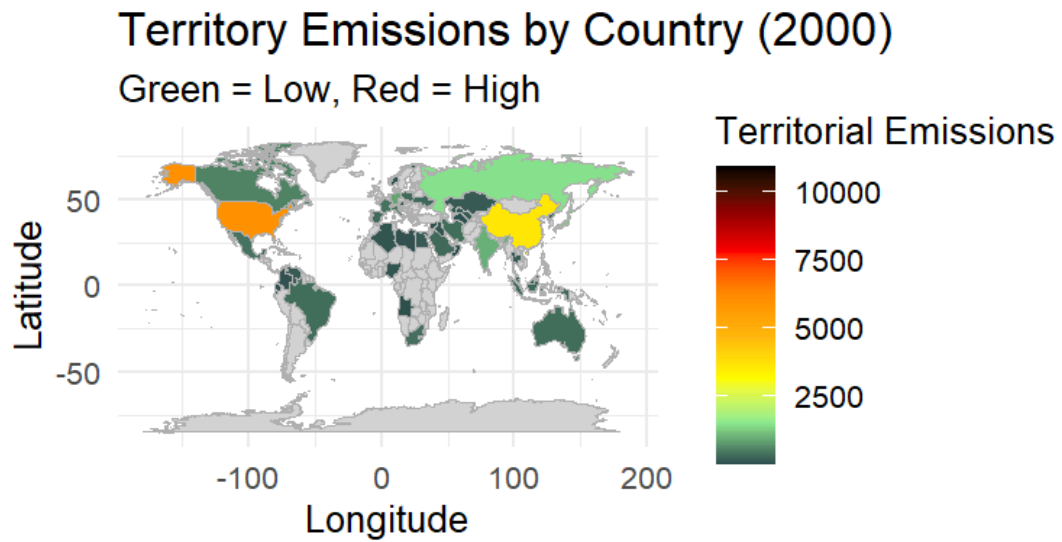


Figure 2.4

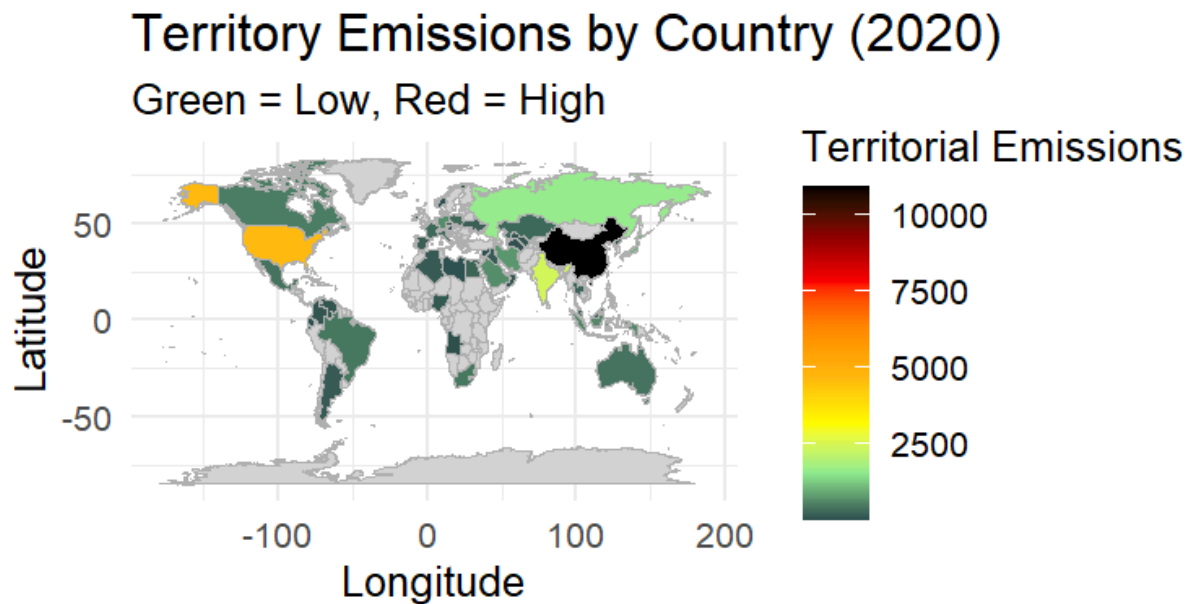


Figure 2 - Production and Territory Comparing Global Distribution from 2000 to 2020

A simple linear regression of corporation count on production emissions revealed a significant positive relationship ($p < 0.001$), with each additional corporation associated with an estimated increase of approximately 203 MtCO₂ in production emissions. The model explained 76.4% of the variation in production emissions ($R^2 = 0.764$), indicating that approximately 76.4% of the variation in production emissions can be explained by the number of corporations alone.

4.2 Global Distribution of Corporate Attribution Ratio Results

The Corporate Attribution Ratio is concentrated at moderately low values, with most countries falling between 0.5 and 4 (Figures 3.1–3.2). Countries at the lower end include Ireland, South Africa, Japan, Spain, and Germany (extremely low), and Egypt, the United States, Poland, Brazil, and India (very low). Mid-range values are represented by Colombia, France, Australia, and Oman (low), and the United Kingdom, Iraq, and Venezuela (moderate). Higher ratios are observed in Kuwait, Qatar, Libya, and

Algeria (high), Switzerland and Angola (very high), and Luxembourg and Norway (extremely high). Some countries, such as Angola, shifted across categories over the study period.

Figure 3.1

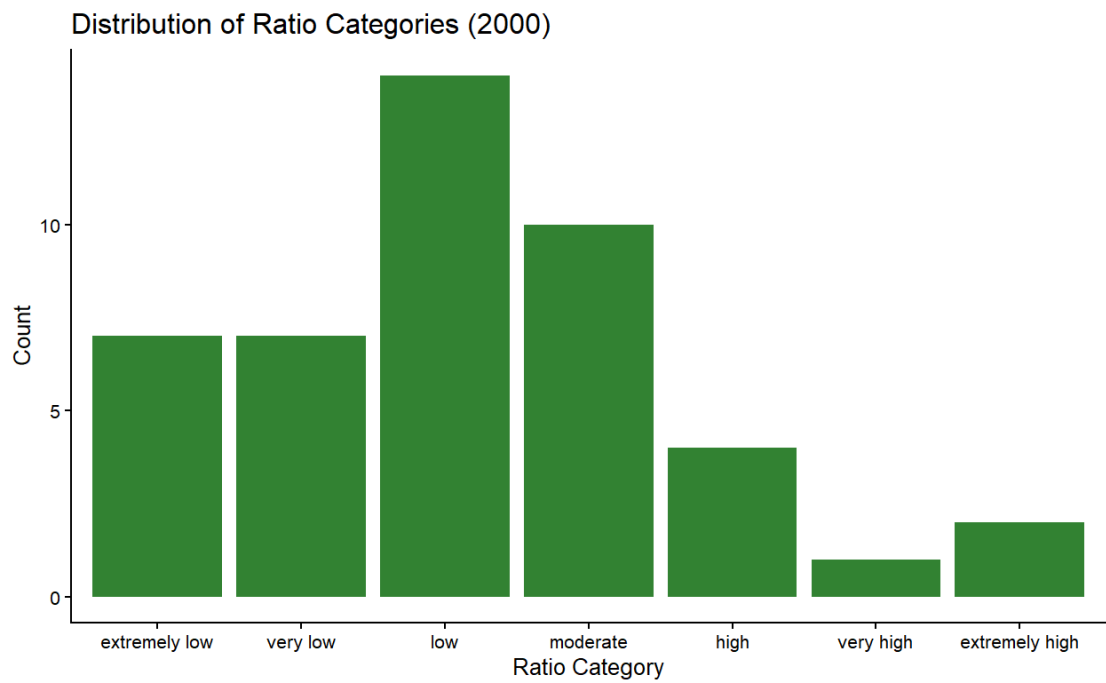


Figure 3.2

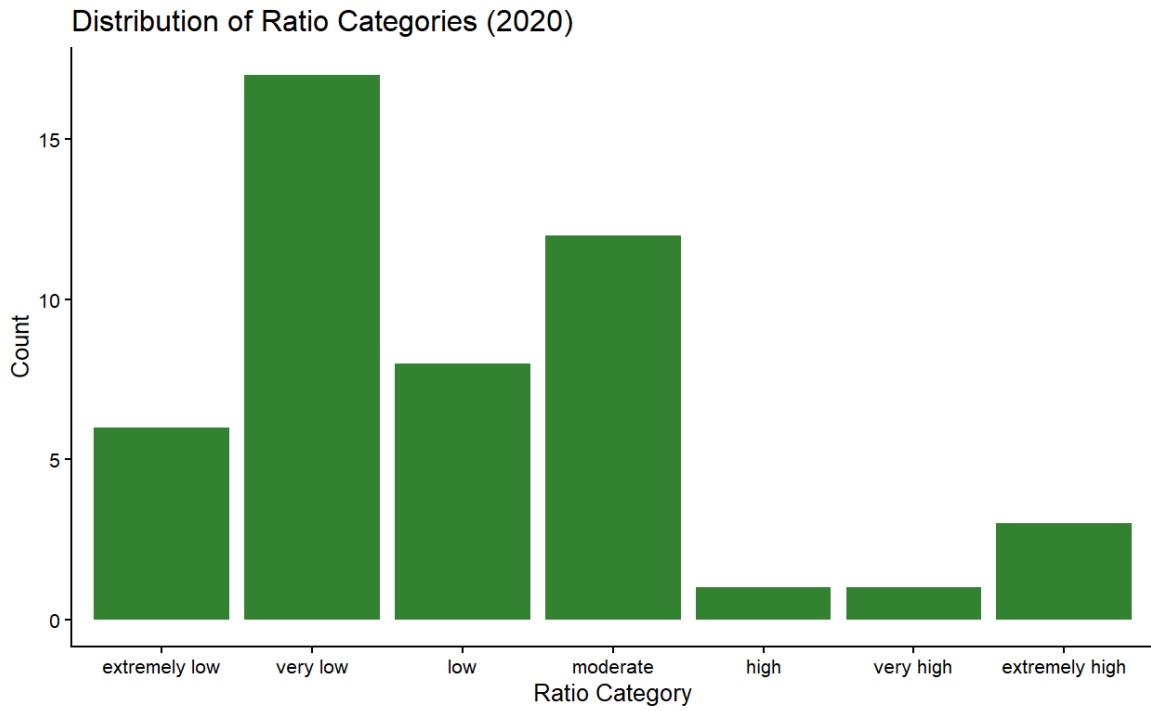


Figure 3 - Bar Chart of Ratio Categories in 2000 and 2020

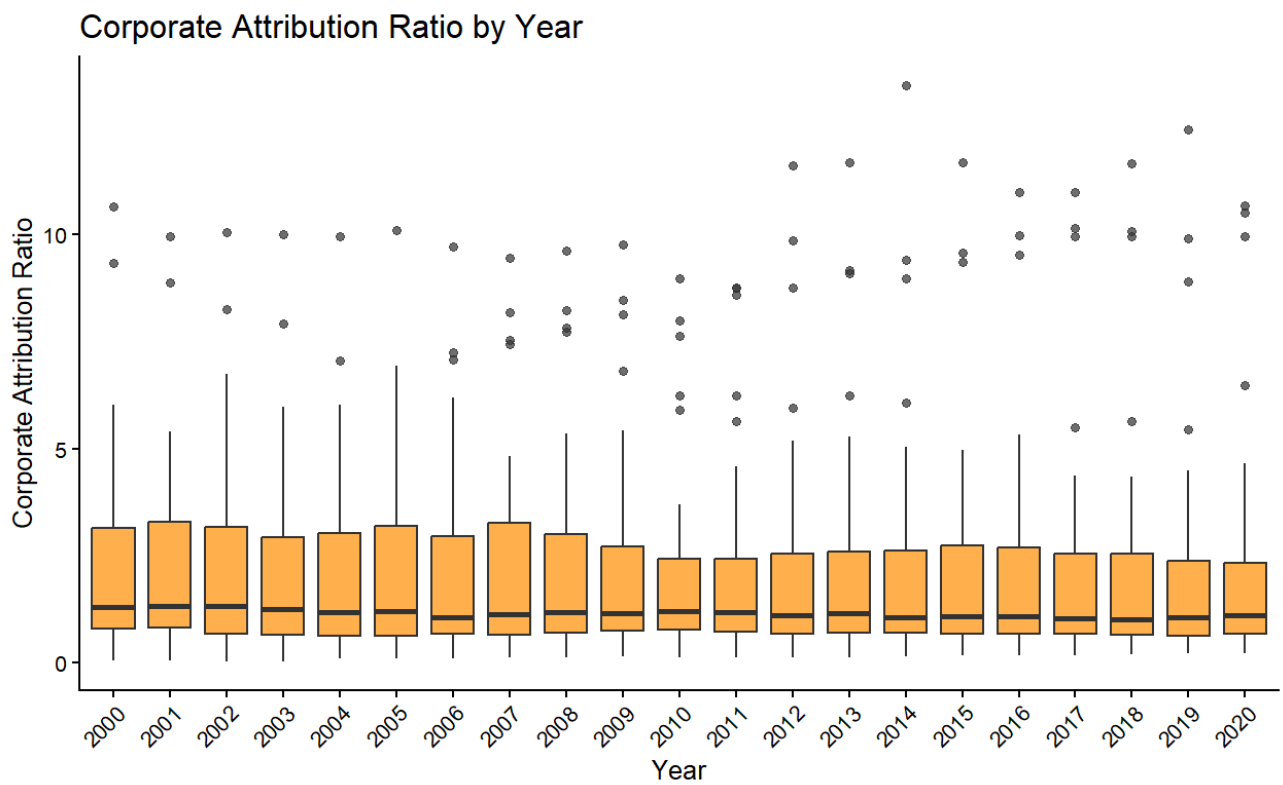


Figure 4 - Side-by-Side Box Plot of Corporate Attribution Ratio from 2000 to 2020

A chi-squared test of independence (with simulation recommended for tables larger than 2×2) found a significant association between continent and categorical Corporate Attribution Ratio ($p < 0.001$). Extremely high ratio values are exclusive to European countries, while Oceania is represented only by low and very low values (Figure 5). Overall, the Corporate Attribution Ratio has a mean of 2.11, a minimum of 0.032, and a maximum of 13.49 (Figure 4).

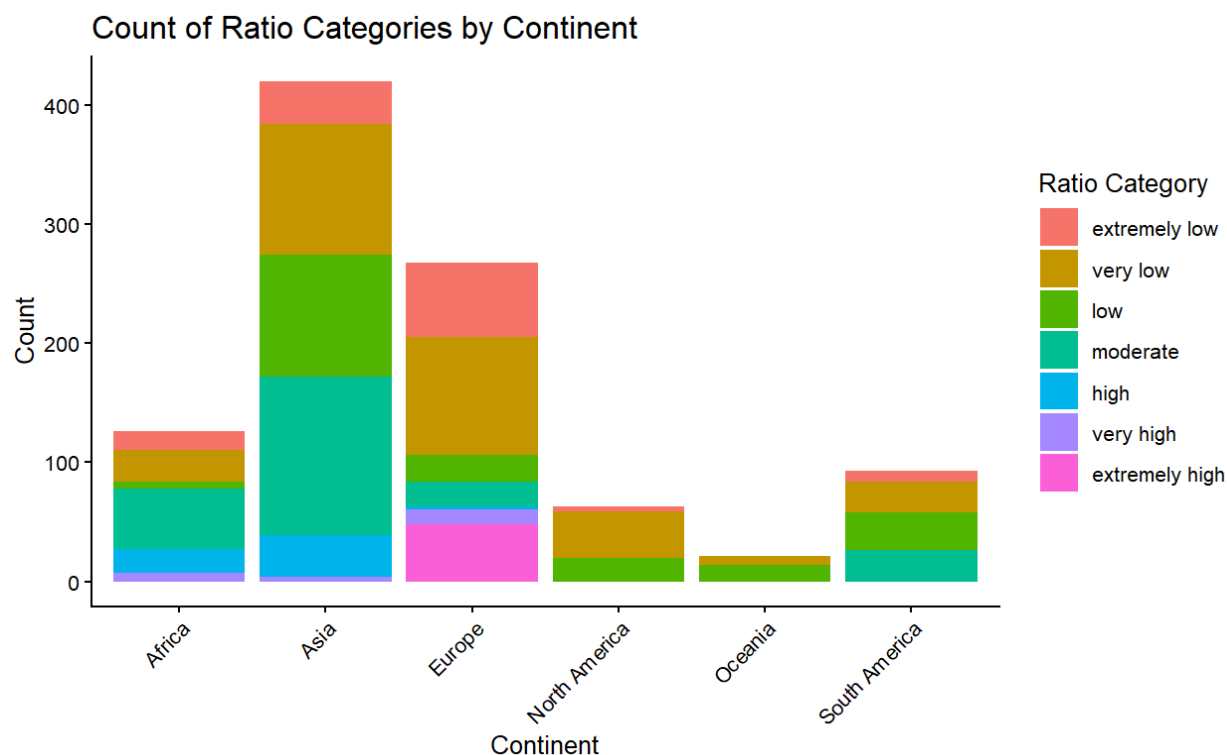


Figure 5 - Categorical Ratio Across Continents Accounting for All Years

Global map visualizations reveal that high production emitters such as the United States and China have relatively low Corporate Attribution Ratios, while low production emitters such as Luxembourg have extremely high ratios (Figures 6.1–6.5). A simple linear regression found a statistically significant positive relationship between production emissions and the Corporate Attribution Ratio ($p < 0.001$); however, the model's low R^2 (0.003) indicates that production emissions alone explain very little of the variation in the ratio.

Figure 6.1

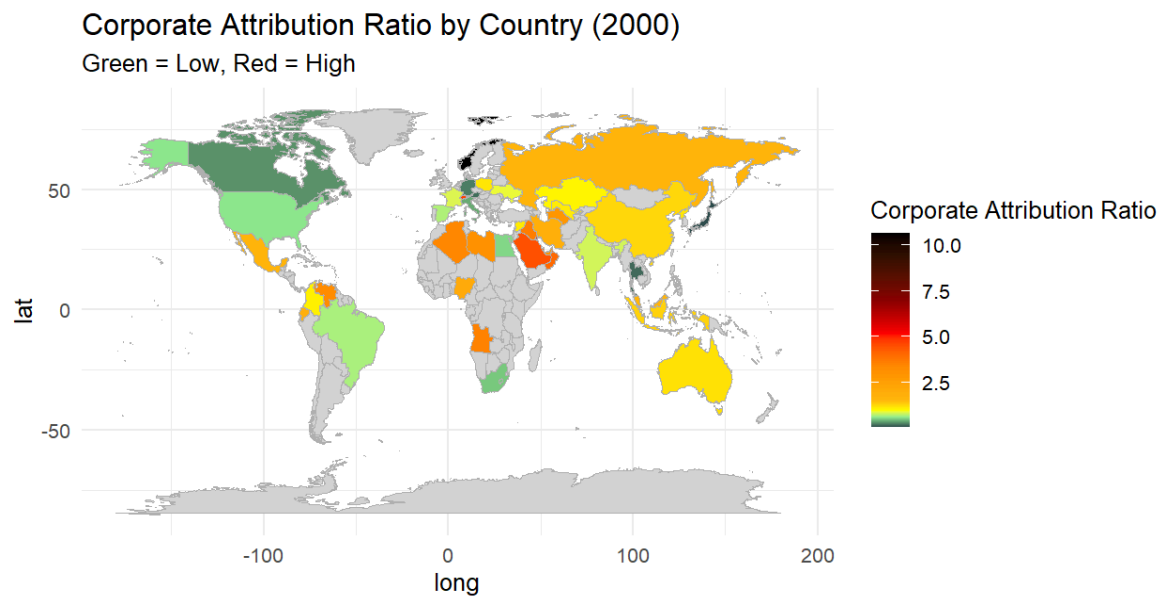


Figure 6.2

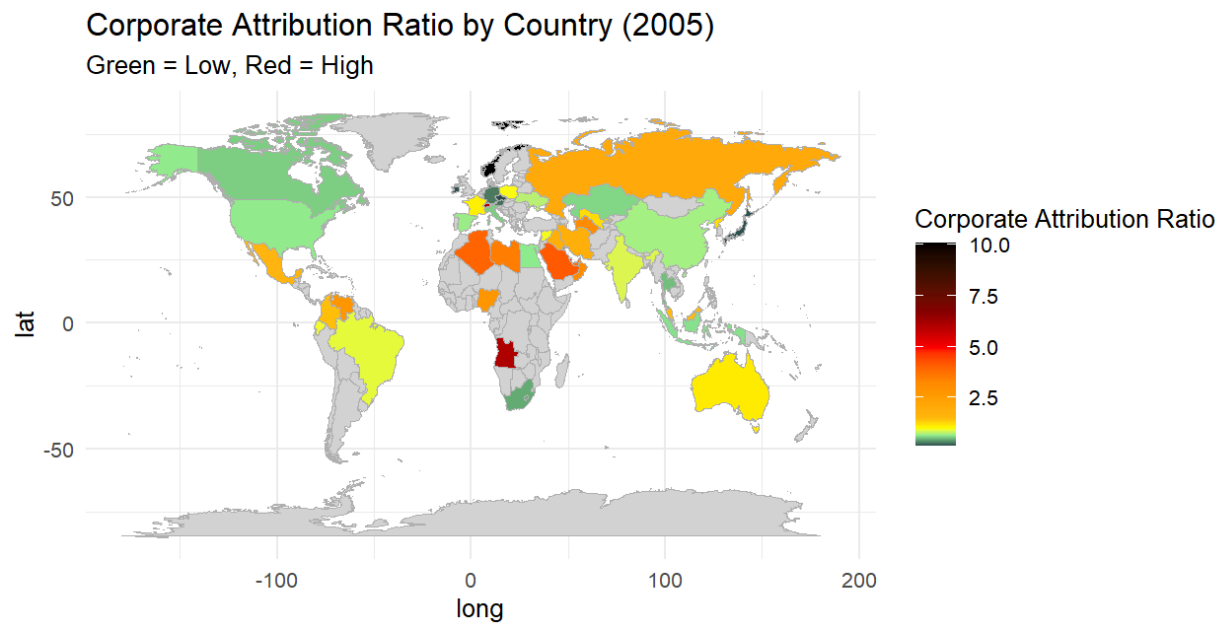


Figure 6.3

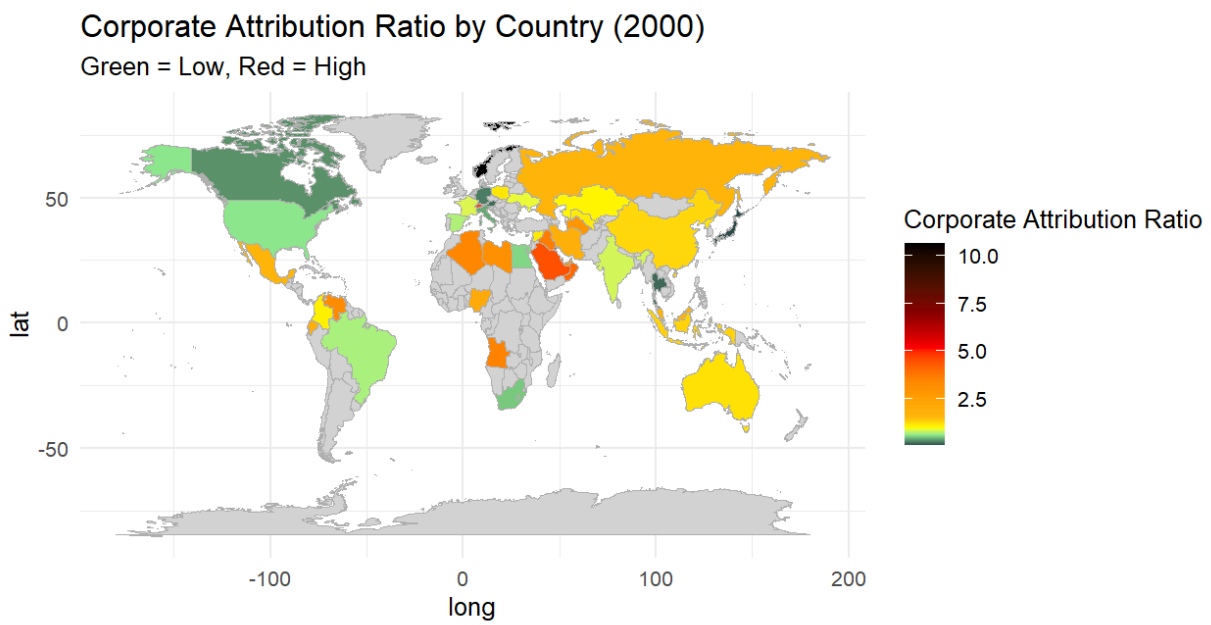


Figure 6.4

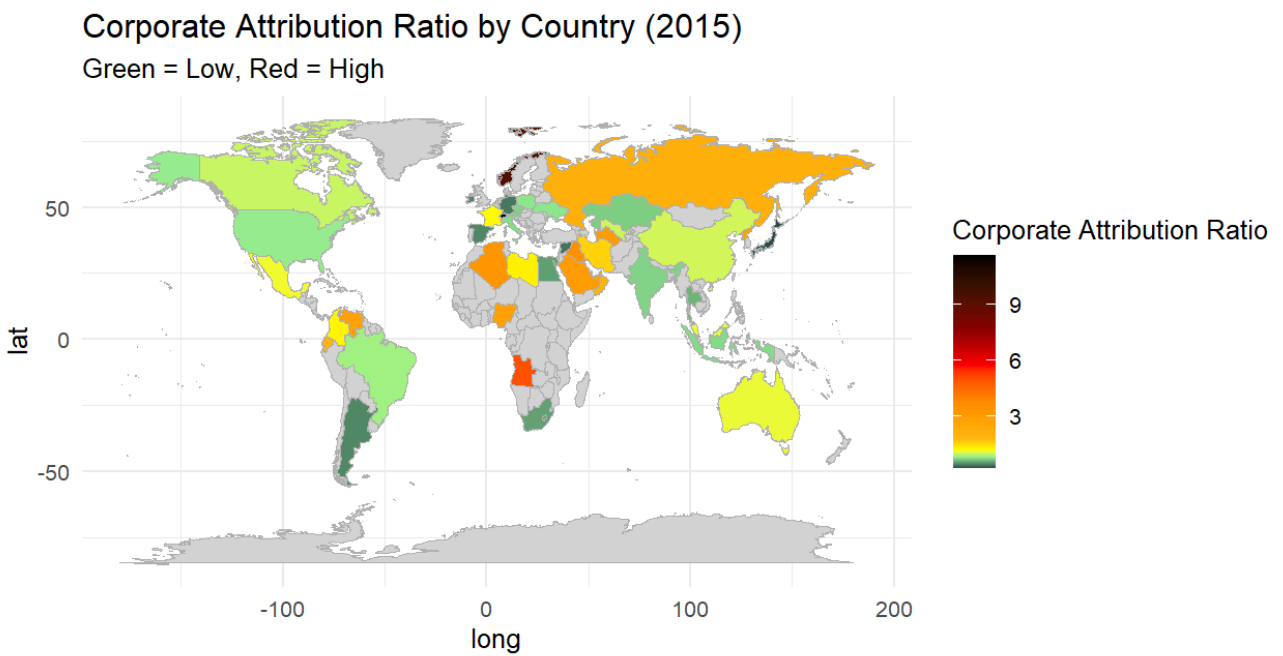


Figure 6.5

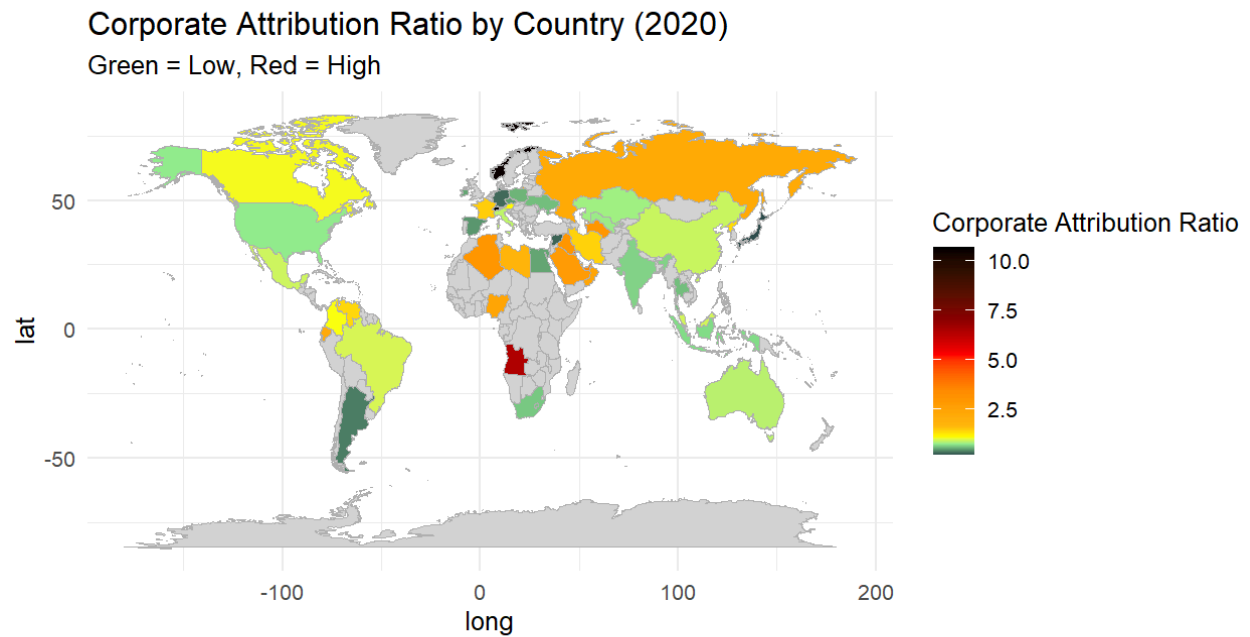


Figure 6 - Corporate Attribution Ratio Global Distribution from 2000 to 2020

4.3 Corporate Attribution Ratio with Time Results

Results showed considerable variation in how the Corporate Attribution Ratio changed from 2000 to 2020. Seventeen countries had statistically significant positive slopes, and sixteen had significant negative slopes, while fourteen showed no significant trend, suggesting stable ratios over the period. Switzerland had the strongest positive trend, followed by Luxembourg, Austria, and Canada. Oman had the strongest negative trend, followed by Libya, Saudi Arabia, Algeria, and Syria. Overall, the Corporate Attribution Ratio did not follow a single global pattern, varying considerably across national contexts.

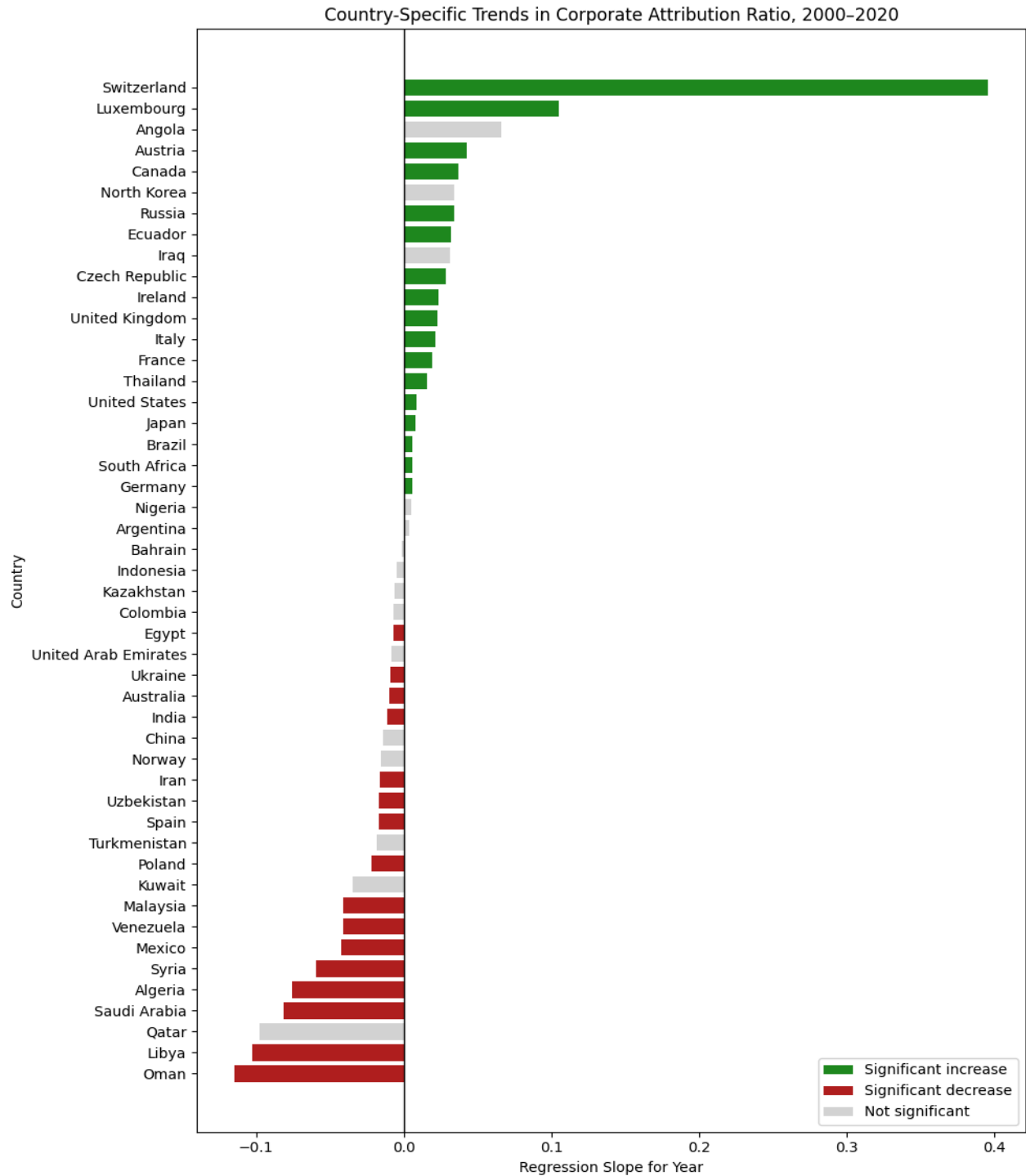
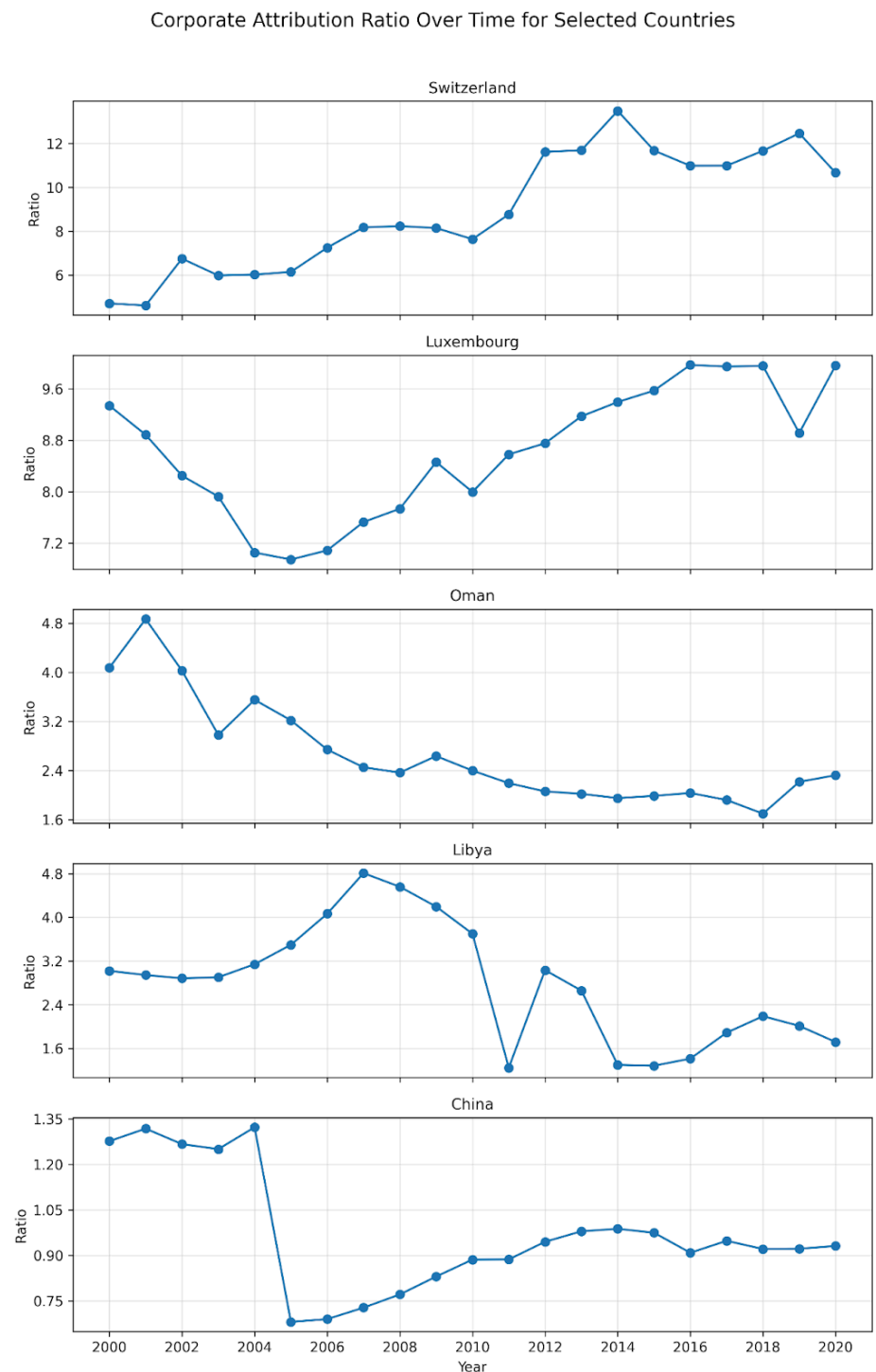


Figure 7 - Country-specific time trends in the Corporate Attribution Ratio from 2000 to 2020.

As shown in Figure 7, the magnitude and direction of the estimated regression slopes varied widely across countries. Figure 8 further illustrates these differences by plotting the annual Corporate Attribution Ratio for selected countries that represent the main patterns observed in the regression results, including

Switzerland, Luxembourg, Oman, Libya, and China. Switzerland and Luxembourg showed overall upward trends, although Luxembourg declined early in the period before increasing later. Oman and Libya displayed overall declines, with Libya showing more pronounced year-to-year variation. China showed a comparatively flatter pattern, consistent with its non-significant regression result. Taken together, these figures suggest that changes in the Corporate Attribution Ratio were highly country-specific over time.

Figure 8 - Corporate attribution ratio over time for selected countries, 2000–2020.



4.4 Difference in Production and Territorial Emissions Results

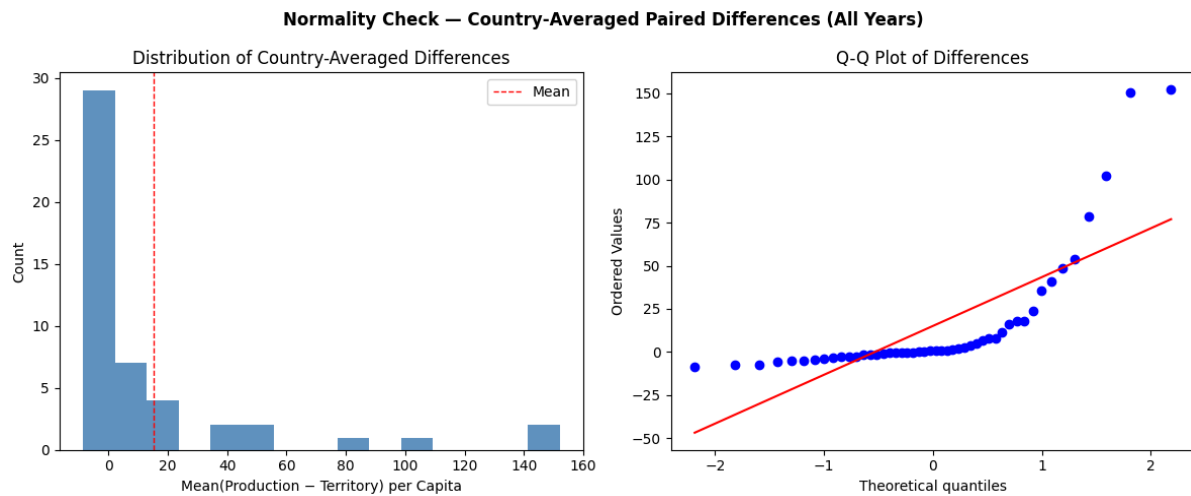


Figure 9 - Heavily right-skewed plot violates normality assumption, cannot trust the paired t-test, but if we did, we would reject the null hypothesis.

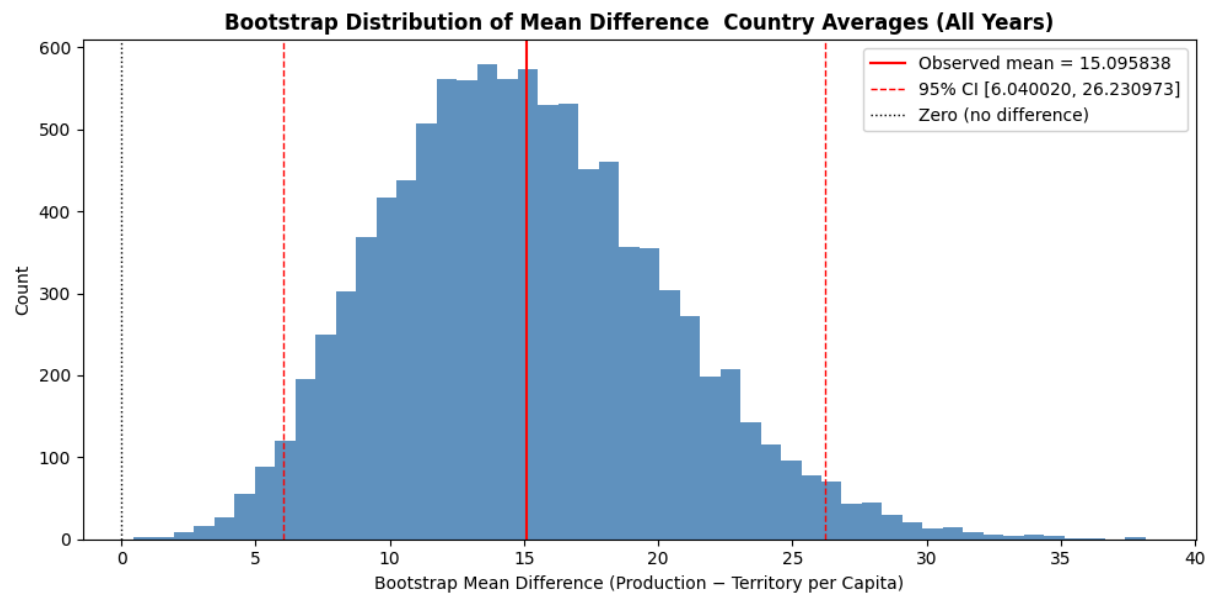


Figure 10 - Bootstrap distribution of the mean difference between production-based and territory-based emissions per capita across countries (2000–2020).

The distribution of country-averaged differences between production and territorial emissions per capita was right-skewed, violating the normality assumption required for a paired test (Figure 9; Shapiro-Wilk:

$W = 0.585$, $p < 0.001$). The Wilcoxon signed-rank test indicated a borderline significant difference ($W = 391.00$, $p = 0.043$). Bootstrap analysis corroborated this finding, yielding an observed mean difference of 15.10 MtCO₂ with a 95% confidence interval of [6.04, 26.23] (Figure 10). Both tests are consistent in suggesting a statistically significant difference between production-based and territorial emissions per capita.

4.5 Corporate Attribution Ratio & Renewable Energy Scores Results (2010 - 2020)

Pearson correlation indicated a weak negative linear relationship between the Corporate Attribution Ratio and renewable energy score ($r = -0.106$, $p = 0.028$), while Spearman correlation revealed a stronger inverse monotonic association ($\rho = -0.408$, $p < 0.001$). The discrepancy suggests the relationship is nonlinear and sensitive to outliers.

A Welch two-sample t-test found that countries where production-based emissions exceeded territorial emissions (ratio > 1) had significantly lower mean renewable energy scores (22.21) compared to those where they did not (38.32; $p < 0.001$). Logistic regression corroborated this, showing that higher renewable energy scores are associated with a significantly lower probability that corporate emissions exceed territorial emissions ($\beta = -0.038$, $p < 0.001$).

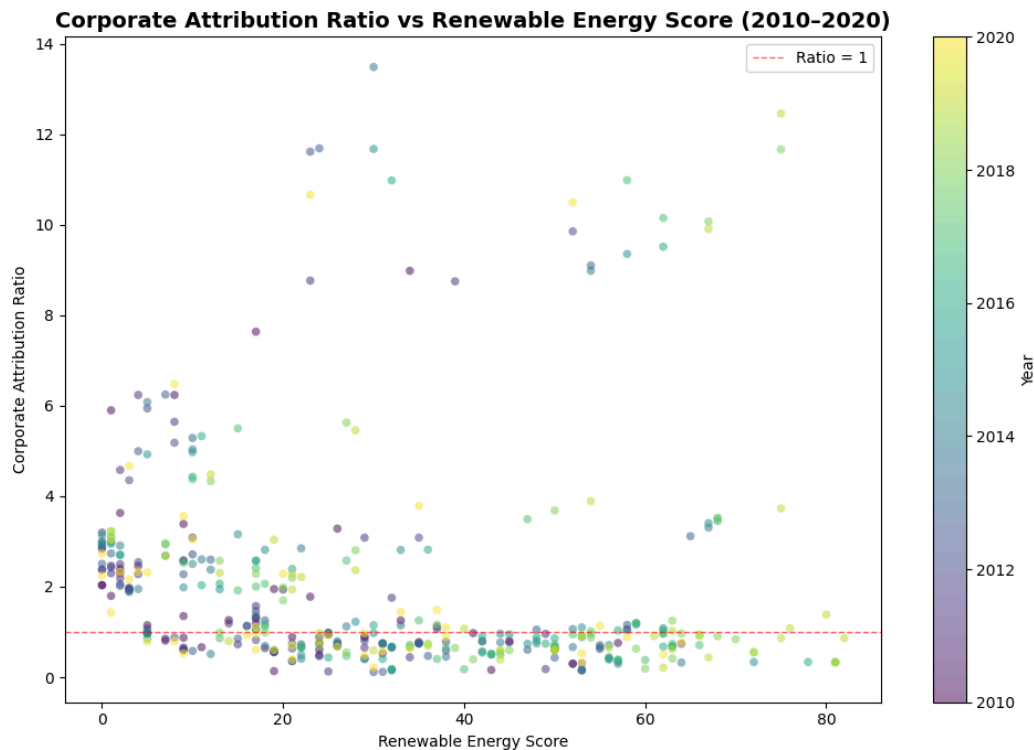


Figure 11 - Scatterplot of Corporate Attribution Ratio vs Renewable Energy Score from 2010 to 2020, color-coded by year. Illustrates the relationship between the Corporate Attribution Ratio and the renewable energy score. The overall pattern is negative, with countries that have lower renewable energy scores exhibiting a wider spread of Corporate Attribution Ratios. Several outliers are observed at high ratio values, particularly among countries with weaker renewable energy policies.

Countries with higher renewable energy scores are more tightly clustered below the ratio = 1 threshold. No clear temporal trend is evident from the year color-coding, suggesting the relationship remained relatively stable from 2010 to 2020.

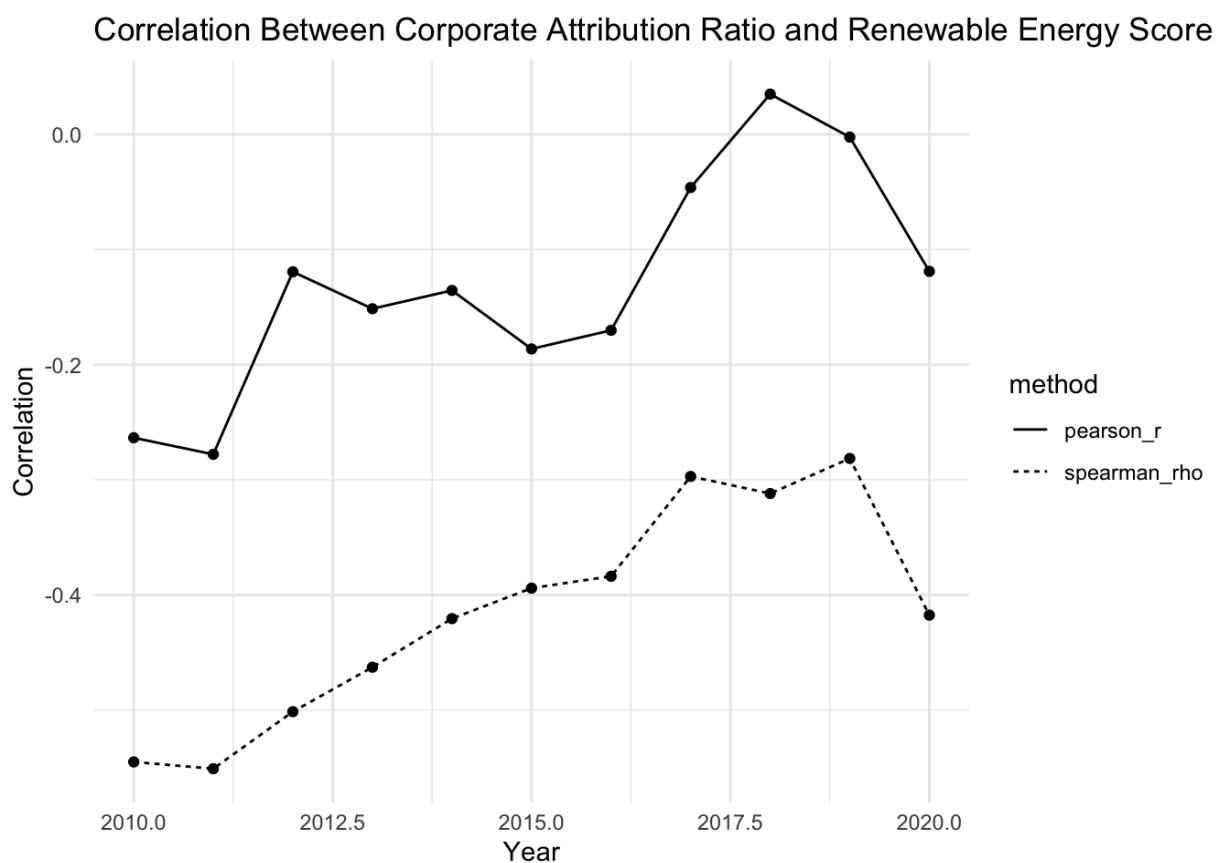


Figure 12 - Yearly Pearson and Spearman correlations between the Corporate Attribution Ratio and renewable energy score, 2010–2020.

Figure 12 displays yearly Pearson and Spearman correlations between the Corporate Attribution Ratio and renewable energy score. Both measures are negative across all years, reinforcing the overall inverse relationship. The Pearson correlation remains weak and fluctuates over time, becoming slightly positive in later years, while the Spearman correlation is consistently negative and of greater magnitude, indicating

a more stable monotonic relationship. The Spearman correlation weakens from 2010 through approximately 2018 before declining again in 2020. The divergence between the two measures suggests the relationship is better characterized as monotonic rather than linear.

4.6 Principal Component Analysis

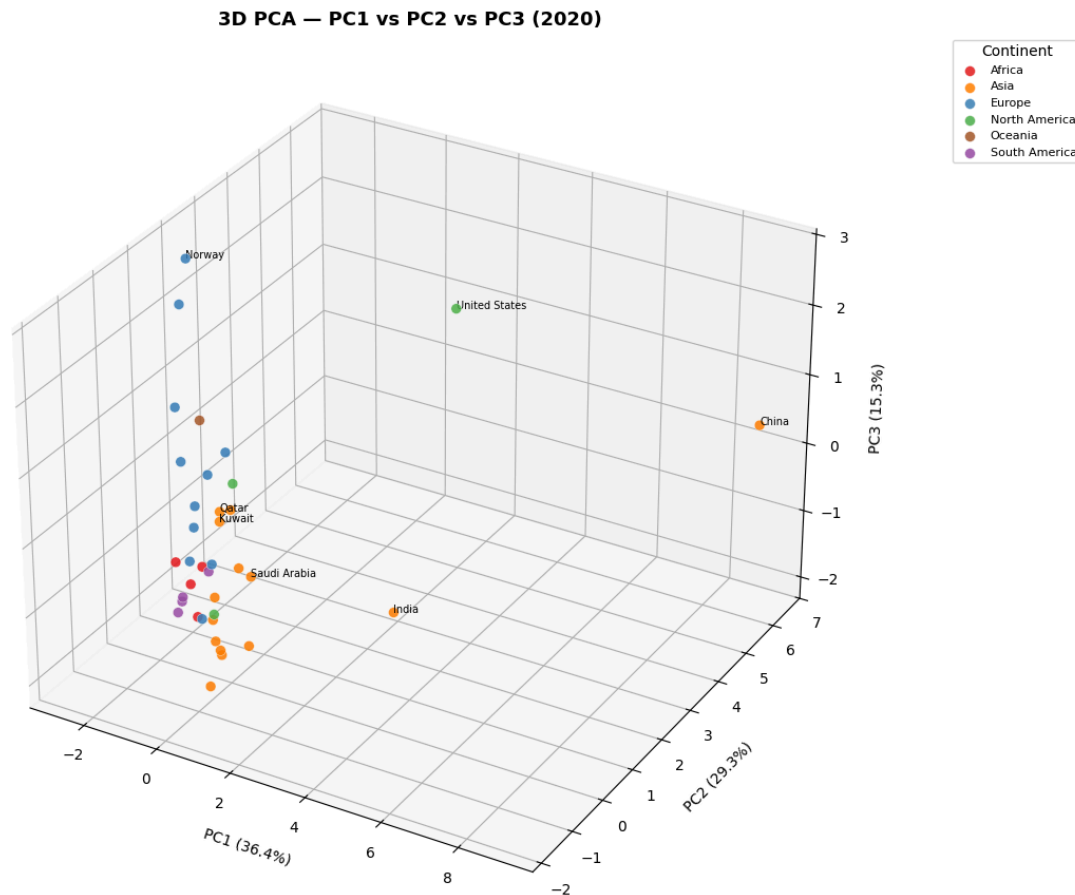


Figure 13 - 3D Scatterplot of PC scores for 2020

Three principal components together explained 81% of the variance (PC1: 36.4%, PC2: 29.3%, PC3: 15.3%); two components were deemed insufficient at 65%. PC1 loaded strongly on production and territorial emissions, population, and corporate count ratio, capturing overall emissions scale. PC2 loaded on per capita production and territorial emissions and their difference, reflecting a per capita rather than absolute emissions dimension. PC3 is loaded primarily on renewable energy score and GDP per capita, separating countries by economic development and clean energy adoption.

As shown in Figure 13, China and the United States were the dominant outliers on PC1, consistent with their status as the largest absolute emitters. Gulf states such as Qatar and Kuwait scored high on PC2,

reflecting small populations but disproportionately high per capita emissions. Norway and the United States were the primary separators on PC3, distinguished by stronger renewable energy and GDP profiles relative to other high-emitting nations.

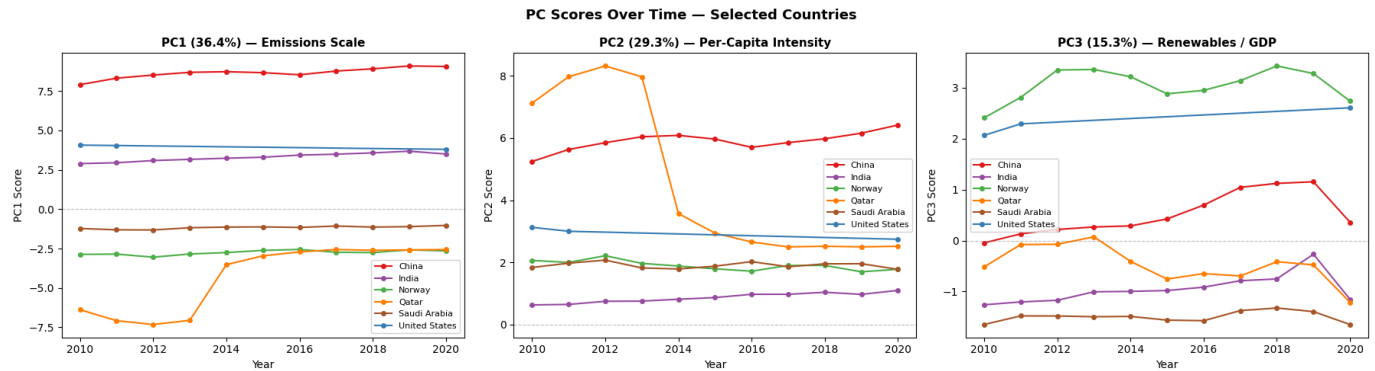


Figure 14 - Longitudinal plot of PC scores from 2010 to 2020 for Select Countries Representative of Each Principal Component

Figure 14 displays principal component scores over time for selected countries. On PC1, China shows a steady increase in overall emissions scale while the United States remains relatively stable at a high level; India also scores high with a gradual upward trend. On PC2, China shows a steady increase in per capita emissions, while Qatar exhibits a sharp decline between 2012 and 2014; all other displayed countries remain stable. On PC3, Norway scores consistently high but with slight fluctuation, and the United States maintains a stable high score throughout the decade.

4.7 Corporate Attribution Linear Regression via Principal Components

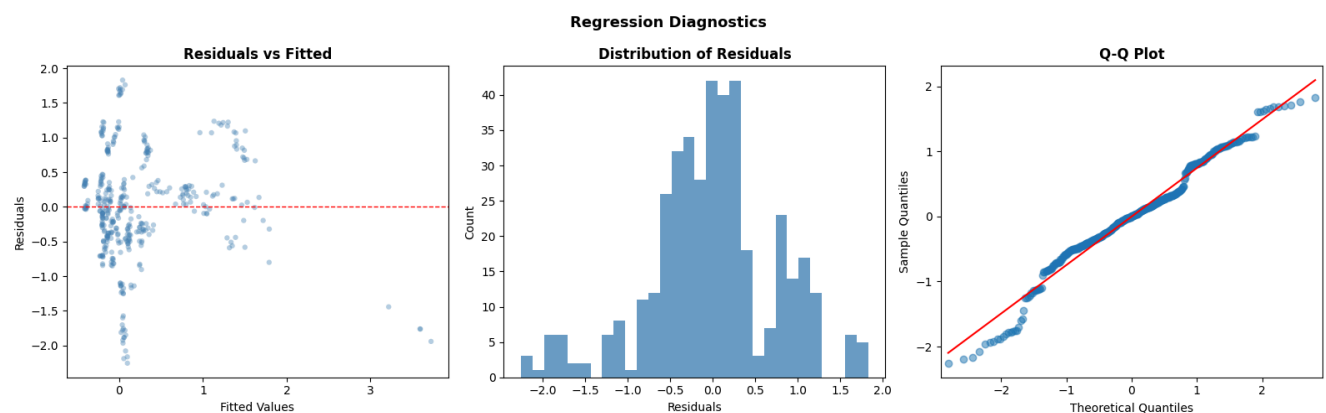


Figure 15 - Regression Diagnostics for $\log(\text{Corporate Attribution Ratio}) \sim \text{PC1} + \text{PC2} + \text{PC3}$

As shown in Figure 15, regression diagnostics show that most model assumptions were substantially satisfied. Residuals vs fitted plot points are mostly scattered randomly around zero, though some concentrated at lower fitted values. Overall the model was statistically significant with an R^2 of 0.4, the 3 Principal Components explain about 40% of the variation in the log corporate attribution ratio. All 3 predictors were significant at $\alpha = 0.05$. PC1 had a coefficient of -0.23, countries with larger absolute emissions tend to have lower corporate attribution ratios. Their territorial emissions are proportionally closer to their production emissions. PC2 had a coefficient of 0.22, countries with disproportionately high per-capita emissions tend to have higher corporate attribution ratios, consistent with what we saw in the PCA analysis graph with Gulf states. PC3 also had a positive coefficient of 0.06, and although it was the weakest predictor of the 3, it still had a p value of 0.025. This suggests that wealthier nations with stronger renewable energy profiles often have higher corporate attribution ratios. We ended up using 3 PCs because with only 2, only 65% of the variance of log corporate attribution ratio is explained.

5.0 Discussion

This report furthered the studies of the Carbon Majors dataset and the Global Carbon Project by comparing production and territorial emissions across countries and years. Production emissions reflect corporate-attributed responsibility linked to fossil fuel and cement producers, while territorial emissions reflect nationally occurring emissions within a country's borders.

Results suggest that both the number of major emitting corporations and their global impact have increased over time. The statistically significant positive trend in total corporations suggests that corporate participation in fossil fuel and cement production has expanded, increasing the complexity of attributing emissions responsibilities across countries. However, this growth is geographically concentrated; the United States, China, and Russia consistently house the majority of corporations, consistent with Heede's finding that a relatively small number of producers drive the majority of global emissions (Heede, 2013; Carbon Majors, 2024).

With the increase in corporations, global emissions have grown, and the widening range of both production and territorial emissions highlights growing inequalities between countries. The strong positive relationship between corporate count and production emissions further supports that corporate presence is a major structural driver of national emissions.

The Corporate Attribution Ratio results reveal an important imbalance: countries with high total production emissions, like the United States and China, have relatively low ratios, while smaller countries

like Luxembourg and Norway exhibit extremely high ratios. This suggests that large economies distribute emissions across many sectors, whereas smaller economies may have emissions dominated by a few major corporations. The chi-squared test confirmed that continent is strongly associated with attribution ratio differences, reflecting regional variation in industrial scale, economic structure, and policy environments. Country-specific time regressions further show that the ratio evolved differently across countries from 2000 to 2020, with no single global trend.

Overall, renewable energy policy showed a more consistent association with the Corporate Attribution Ratio than GDP per capita. Pearson and Spearman correlations between GDP per capita and the ratio were small and not statistically significant, suggesting that economic development alone does not explain differences in corporate emissions profiles. In contrast, the Spearman correlation between renewable energy score and the ratio was stronger and statistically significant, holding across both pooled and year-specific analyses. The stronger Spearman relative to Pearson results indicate the relationship is monotonic rather than strictly linear, likely influenced by a small number of extreme observations consistent with the skew identified earlier in the analysis.

The principal component analysis provided a complementary perspective on the structure of emissions-related variables across countries. Three components explained 81% of the total variance, capturing distinct dimensions of emissions behavior. PC1, which loaded strongly on absolute production and territorial emissions, population, and corporate count, reflects overall emissions scale, consistent with earlier findings that a small number of large economies dominate global emissions volumes. PC2, defined by per capita production and territorial emissions and their difference, separates countries by emissions intensity relative to population size, highlighting that some smaller nations, particularly Gulf states like Qatar and Kuwait, carry disproportionately large per capita emissions footprints despite low absolute totals. PC3, driven by renewable energy score and GDP per capita, captures economic development and clean energy adoption, distinguishing countries like Norway and the United States from high-emitting nations with weaker policy profiles.

The temporal trajectories of PC scores further reinforce several key findings. China's steady increase on both PC1 and PC2 reflects its growing dominance in both absolute and per capita emissions over the study period, while the United States remains stably high across components. Qatar's sharp decline on PC2 between 2012 and 2014 suggests a notable shift in per capita emissions intensity during that period, warranting further investigation. Together, the PCA results confirm that no single dimension fully captures cross-country variation in emissions patterns, and that scale, intensity, and policy environment represent meaningfully distinct axes of differentiation.

There are several limitations to this analysis. The RISE renewable energy score measures policy design rather than implementation or effectiveness, meaning high scores do not guarantee high renewable adoption. The analysis is also limited to countries and years available across the combined datasets. Most importantly, corporate-attributed emissions are assigned based on firm headquarters rather than where emissions physically occur, meaning the Corporate Attribution Ratio reflects global corporate influence rather than domestic environmental impact, particularly for countries hosting large multinational fossil fuel firms. Additionally, because Scope 3 emissions capture downstream global use of sold products, the ratio may systematically overstate production-based responsibility relative to territorial emissions for major exporting nations.

Future research could incorporate actual renewable energy usage metrics, apply panel regression or mixed-effects models to account for repeated country observations, and explore regional or development-status groupings for additional insight.

Overall, this study highlights the importance of distinguishing between production-based and territorial emissions when evaluating global carbon responsibility. Policy-related factors, particularly renewable energy regulation, show a more consistent relationship with corporate emissions patterns than economic development alone.

6.0 Conclusion

This study highlights the importance of distinguishing between production-attributed and territorial carbon emissions when evaluating global fossil fuel CO₂ responsibility. By constructing the Corporate Attribution Ratio across countries from 2000 to 2020, we showed that corporate-linked emissions can differ substantially from territorial emissions and that this relationship varies meaningfully across countries and over time. Large economies such as the United States and China distribute emissions across many sectors, yielding relatively low ratios, while smaller countries are often dominated by fewer firms, producing disproportionately high ratios.

GDP per capita showed little explanatory power over the Corporate Attribution Ratio, while renewable energy regulatory strength demonstrated a more consistent association with lower ratios and a reduced likelihood that production-attributed emissions exceed territorial emissions. This pattern was stronger in rank-based analyses than linear ones, suggesting the relationship is monotonic and sensitive to outliers. Together, these findings suggest that policy context may matter more than economic development in shaping how corporate emissions relate to national emissions.

Several limitations should be noted. The Corporate Attribution Ratio relies on estimated production-linked emissions attributed by headquarters location rather than where emissions physically occur, which may overstate corporate responsibility for major exporting nations. The RISE score reflects policy design rather than implementation or outcomes. Additionally, 2020 data may be affected by the COVID-19 pandemic, during which widespread economic shutdowns caused sharp declines in emissions that could distort longer-term trajectories. Future work could incorporate actual renewable energy usage metrics, expand country coverage, apply longitudinal modeling techniques, and examine counterfactual emissions trends absent the pandemic's disruption.

Overall, this study confirms that corporate activity plays a central and increasingly influential role in global carbon emissions patterns and that environmental policy frameworks are more closely associated with corporate emissions behavior than economic development alone. The Corporate Attribution Ratio offers a useful and replicable framework for future research comparing corporate and territorial perspectives on carbon responsibility.

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8.0 Author Contributions

Jacob Namath

Responsible for data curation and cleaning, including merging 3 of the datasets to get a broader look at all of the information necessary. Conducted statistical analyses for 3.5, 3.6, and 3.7, 3.8. These encompassed the comparison of production and territorial fossil CO₂ emissions per capita in 2020, the analysis of Corporate Attribution Ratio associations with GDP per capita from 2000 to 2020 across countries, and the

association between corporate attribution and renewable energy regulations from 2010 to 2020. Also performed Principal Component Analysis and built a linear regression model based on those components to predict Corporate Attribution Ratio.

Casey Huczko

Completed preliminary analysis regarding production emissions, the number of corporations, and the initialization of the categorical ratio variable. Analyzed the relationship between continent and categorical ratio to test initial findings. Directed the analysis of the Corporate Attribution Ratio utilizing “maps” in R, such as cleaning data to ensure all relevant countries were represented, enhancing the scale to represent all ratio values, and analyzing the visuals. Completed preprint editing, formatting, and organization.

Brenna Henry

Assisted with data cleaning, including dropping unnecessary values, relabeling columns, and preparing the final dataset used in the study. Conducted the country-specific analysis of Corporate Attribution Ratio trends from 2000 to 2020 by implementing separate ordinary least squares regressions for each country in Python, extracting slope, p-value, and R^2 estimates, and classifying trends as positive, negative, or non-significant. Created the corresponding visualizations used to summarize country-specific regression slopes and illustrate selected time trends.

Katy Waterman

Led the integration and analysis of the RISE renewable energy dataset, including variable selection, interpretation of policy indicators, and alignment with the compiled dataset. Contributed to the statistical analysis examining the relationship between renewable energy scores and the Corporate Attribution Ratio, including correlation analyses and regression modeling. Assisted with the interpretation of results and contributed to writing and revising the methods, results, and discussion sections.

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9.1 AI Disclosure

AI tools were used to assist with code development, data processing, and editing. All analytical decisions, interpretations, and conclusions were made by the authors.